Fiscal Year:	FY 2014	**Task Last Updated:**	FY 11/19/2013
PI Name:	Blakely, Eleanor A Ph.D.		
Project Title:	Harderian Gland Tumorigenesis: Low-Dose-, Low-Dose-Rate-, and LET-Response		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiation NSCOR/Virtual NSCOR NNJ10ZSA002N
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 3/31/2016, per S. Monk/LaRC (Ed., 4/5/16) NOTE: Extended to 12/31/2015, per PI (Ed., 8/13/15) NOTE: Extended to 6/30/2015 per PI (Ed., 10/8/14) NOTE: Gap change to delete Cancer 05, per PI (Ed., 12/17/12)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chang, Polly (SRI International)		
Grant/Contract No.:	NNJ11HA94I		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Increased cancer risk remains a primary concern for travel into deep space and may preclude manned missions to Mars due to large uncertainties that currently exist in estimating cancer risk from the spectrum of radiations found in space with available human epidemiological radiation-induced cancer data. Existing data on human risk of cancer from exposures to X- and gamma-rays must be scaled to the many types and fluences of radiations found in space using radiation quality factors and dose-rate modification factors, and assuming linearity of response since the shapes of the dose-responses at low doses below 100 mSv are unknown. The goal of this proposal is to reduce uncertainties in the RBE/LET relationship for particle radiation carcinogenesis. Our approach is to use the existing data on Harderian gland tumorigenesis as reference, and obtain Harderian gland tumor data for heavy ions beams not previously reported to fill the gaps in the LET range from below 25 keV/micrometer and up to approximately 193 keV/micrometer to improve our understanding of the dose-response curve at low doses and to test for deviations from linearity. The hypothesis to be tested is that filling in these data gaps will delineate more comprehensively the space radiation risk of space radiation induced tumorigenesis, in particular to provide data that will allow us to test for a role of targeted versus non-targeted effects in cancer risk. We aim to investigate the effects of a broader range of radiation qualities, doses and dose rates not previously reported on the development of Harderian gland and also lung tumors in ~100-120 day old female CB6F1 mice 16 months after exposure.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	The successful completion of the proposed work will fill a significant gap in Harderian tumor response data at low fluences of Silicon or Titanium particle beams to fill in the tumor prevalence in the LET range from below 25 keV/micrometer and up to approximately 193 keV/micrometer to improve our understanding of the dose-response curve at low doses relevant to space travel and to test for deviations from linearity. The data should contribute to decreasing the current uncertainty in estimating radiation risk of cancer due to exposures in space travel.		
	We are on schedule with our proposed animal irradiations and cohort sacrifices for these large animal studies required for the statistical power to achieve our goals of documenting prevalence of carcinogenesis after low dose/low fluence exposures of specific high-energy, space-relevant charged particles at relatively high-, as well as low-dose rates. Our focus is to investigate non-targeted effects at particle fluences of high atomic number particles relevant to human exposures in space. This animal model was selected to replicate the choice of an animal with a low background of Harderian Gland tumor incidence, and an extensive published data-base of tumorigenesis with numerous particle beams, but lacking data between 25 keV/micrometer and 193 keV/micrometer. During the 2013 funding period four separate purpose-bred female B6CF1 (F1 of C57Bl6 X BALB/c) female murine cohorts were irradiated at between 100-120 days old. These included: 260 mice that were sham-irradiated or exposed to single acute doses of either 0, 3, 6.5, 13, 26, or 52 cGy of 1 GeV/u Titanium ions. In parallel, 240 additional mice received five daily dose fractions of either 2.6 cGy or 5.2 cGy. A separate cohort of 180 mice were exposed to single acute dose fractions of 0, 10, 20, or 40 cGy of 600 MeV/u Iron ions. Finally, 500 mice were exposed to a single acute dose of either 0, 40, 80, 120, or 160 cGy of 137Cs gamma rays. These animals are being monitored with careful husbandry since their return to Berkeley Lab, and will be followed for 16 months until their scheduled necropsy. In July/August 2013, we harvested the animal cohorts irradiated at Brookhaven National Laboratory's NASA Space Research Laboratory 16 months prior with 260 MeV/u Silicon ions at 70 keV/micrometer, or 1 GeV/u Titanium ions at 100 keV/micrometer. Complete necropsies were completed and all tumor and tissue samples from numerous organs were harvested and either snap frozen or fixed for comprehensive histology and evaluation by a pathologist. The analysis of our results reveals: (1) The sham-treated, but unirradiated, animals have a low Harderian Gland (HG) tumor incidence that is comparable to previously published control incidence of Alpen et al. 1993, and confirm our replication of the same animal strain and protocols; (2) The dose-dependent HG tumor incidence data following exposure to 260 MeV/u Silicon ions at 70 keV/micrometer lie nested		

Task Progress:	between the published Alpen data of 600 MeV/u Iron ions at 193 keV/micrometer, and the Alpen data of 670 MeV/u Neon ions at 25 keV/micrometer on an tumor incidence versus particle dose plot. The fluence dependence (incidence versus tracks per cell nuclei plot) shows a very flat response until the high dose at 26 cGy. It is important to note that we only found one animal in this study with bilateral HG tumors. The majority of the animals presented with only one HG tumor; however, tumors were found in a number of other tissues. The analysis of other types of tumors following Silicon ion exposures shows a similar early dose response, but a steeper higher dose response followed by what may be saturation. The dose response of the Si-induced lung tumors revealed non-linearity with an early steep dose response, followed by a nadir at 16 cGy, and then an increase at 32 cGy. Colon tumors were evaluated for all of these animals in a collaboration with Dr. Jerry Shay of UTSW in Dallas, TX. Dr. Shay reported absolutely no colon tumors were found in our Si-irradiated animals. The dose response of the Titanium HG incidence at 100 keV/micrometer is very similar to the Si-induced incidence, overlapping at low dose and lying just above the Si-data at the two high dose points. The fluence dependence for Ti-induced HG tumors at 100 keV/micrometer lies between the Fe data at 193 keV/micrometer and the Si data at 70 keV/micrometer. Histology of the HG tumors revealed primarily benign papillary adenomas, with a small percentage of carcinomas in the sham-treated controls and 16 and 32 cGy high dose groups. The histology of the sham-treated control lung tumors revealed equal numbers of primarily bronchio-alveolar adenoma or lymphomas. Si-induced histologies revealed non-linear dose-dependent differences in tumor number and histology. Bronchio-alveolar adenomas were prevalent, with small percentages of bronchio-alveolar adenocarcinoma, histiocytic/reticulum cell carcinoma and lymphoma. Next Summer’s harvest of the remaining four cohorts of single acute doses or 5-fractions of Titanium, the single acute doses of 600 MeV/micrometer Iron, and the 137Cs gamma ray study will complete our experimental program, and allow the comprehensive comparisons of all of the data from our work.
Bibliography Type:	Description: (Last Updated: 02/12/2026)
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